Foundations of AI and Machine Learning - Module 1 Class Notes

Session 1: Introduction to Artificial Intelligence

Sub-session 1.1: Core Concepts of AI

- **Teacher Entry:** "As we embark on our AI journey, let's start by understanding its essence. Welcome to the Core Concepts of AI!"
- **AI (Artificial Intelligence):** Systems designed to mimic human intelligence, including learning, reasoning, and perception.
    - **Explanation:** AI processes data to identify patterns and make decisions, often without explicit programming.
    - **Analogy:** A skilled apprentice learning crafts by observing a master, or a self-driving car adapting to roads.
    - **Relevance to Present:** Underpins ChatGPT (OpenAI) for conversation and Grok (xAI) for insights.
- **Packages:**
    - **Pandas:** Manages data for analysis, essential for loading datasets.
- **Class Work Sub-sub-session 1.1.1: Class Work - Set Up VS Code with Jupyter and Development Environment**
    - **Activity:** Install VS Code, Jupyter extension, Python, and Pandas.
    - **Resources:** VS Code, terminal, text editor.
    - **Step-by-Step Guide:**
        a. Download and install VS Code from code.visualstudio.com.
        b. Open VS Code, go to Extensions (Ctrl+Shift+X or Cmd+Shift+X), search for "Jupyter," install the official Microsoft Jupyter extension.
        c. Open terminal (Ctrl+` or Cmd+`), type `python --version` or `python3 --version`; if no version, visit python.org/downloads, download 3.11, install with "Add Python to PATH."
        d. Verify with `python --version` (e.g., 3.11.0).
        e. Type `pip install pandas` and press Enter; wait for "Successfully installed."
        f. In VS Code, create a new file (Ctrl+N or Cmd+N), save as `Setup.ipynb`, select

Python kernel from the Jupyter toolbar.
    g. In a new cell, type `import pandas as pd; print("Hello AI")`, run with Shift+Enter, expect “Hello AI.”
  - **Analogy:** Setting up a workshop with tools.

Sub-session 1.2: Historical Milestones

- **Teacher Entry:** "With the basics established, let’s travel back to see how AI evolved. Welcome to Historical Milestones!"
- **The Turing Test (1950):** Proposed by Alan Turing to assess machine intelligence via text conversation.
  - **Explanation:** A human judge chats with a machine and human; if the machine deceives, it’s intelligent.
  - **Analogy:** A blind taste test of robot vs. human cooking, or a phone guess game.
  - **Significance:** Shaped benchmarks for ChatGPT and early ELIZA.
- **Deep Blue’s Victory (1997):** IBM’s Deep Blue beat chess champion Garry Kasparov.
  - **Explanation:** Used algorithms to evaluate millions of moves, surpassing human strategy.
  - **Analogy:** A super-fast librarian finding chess moves, or a tireless calculator.
  - **Outcome:** Pioneered AI in strategic fields like logistics.
- **Dawn of Deep Learning (2000s):** Neural networks with multiple layers emerged for pattern recognition.
  - **Explanation:** Mimics brain layers to process data like images or speech.
  - **Analogy:** A student learning animal shapes from a photo album, or a painter layering colors.
  - **Relevance to Present:** Drives Gemini (Google) and Meta AI image tools.
- **Class Work Sub-sub-session 1.2.1: Class Work - Research a Milestone**
  - **Activity:** Research one milestone and summarize its impact.
  - **Resources:** Internet, notebook.
  - **Step-by-Step Guide:**
    a. Open a browser and search “AI historical milestones” (e.g., Turing Test).
    b. Select one milestone (e.g., Deep Blue 1997).
    c. Read about it and note its year, key event, and impact (e.g., “1997, Deep Blue beat Kasparov, advanced strategic AI”).
    d. Write a 2-sentence summary in your notebook.

    e. Share with a partner.
  - **Analogy:** Exploring a history book chapter.

Sub-session 1.3: Types of AI

- **Teacher Entry:** "Having traced AI's history, let's categorize its forms. Welcome to Types of AI!"
- **Narrow AI:** Specialized systems like ChatGPT (conversation), Grok (queries), Gemini (search), Meta AI (content), and predecessors ELIZA (chatbot) and ALVINN (driving).
  - **Explanation:** Limited to specific tasks, unable to generalize.
  - **Analogy:** A one-trick coffee machine, or a locksmith for one lock brand.
- **General AI:** Hypothetical AI performing any human task, envisioned for future Grok or ChatGPT successors.
  - **Explanation:** Versatile across domains.
  - **Analogy:** A Swiss Army knife, or a universal translator.
- **Superintelligent AI:** Hypothetical AI surpassing humans, potentially from Gemini or Meta AI advances.
  - **Explanation:** Solves beyond human capacity, raising ethics.
  - **Analogy:** A cosmic architect, or a super-genius teacher.
- **Class Work Sub-sub-session 1.3.1: Class Work - Match AI Types**
  - **Activity:** Match AI examples to their types.
  - **Resources:** Handout with examples.
  - **Step-by-Step Guide:**
    a. Collect the handout with examples (e.g., Siri, hypothetical super-AI).
    b. Work in pairs as assigned.
    c. Read each example and decide its type (narrow, general, superintelligent).
    d. Write matches on the handout (e.g., "Siri: Narrow AI").
    e. Compare with the teacher's answers.
  - **Analogy:** Sorting tools into a toolbox.

Session 2: Introduction to Machine Learning

Sub-session 2.1: Fundamentals

- **Teacher Entry:** "Building on AI, let's dive into how machines learn. Welcome to the

Fundamentals of Machine Learning!"

- **Machine Learning (ML):** AI subset where systems improve from data, not rules.
  - **Explanation:** Algorithms detect patterns and adjust predictions.
  - **Analogy:** A gardener tweaking soil for better flowers, or a detective using past clues.
  - **Relevance to Present:** Core to ChatGPT and Gemini.
- **Class Work Sub-sub-session 2.1.1: Class Work - Install ML Libraries**
  - **Activity:** Install Scikit-learn and verify in VS Code.
  - **Resources:** Terminal, VS Code.
  - **Step-by-Step Guide:**
    a. Open terminal in VS Code (Ctrl+` or Cmd+`).
    b. Type `pip install scikit-learn` and press Enter; wait for “Successfully installed.”
    c. In `Setup.ipynb`, add a new cell, type `import sklearn; print("Scikit-learn installed")`, run with Shift+Enter, expect “Scikit-learn installed.”
  - **Analogy:** Adding cooking tools to your kitchen.

Sub-session 2.2: Learning Types

- **Teacher Entry:** "With fundamentals clear, let’s explore how ML learns. Welcome to Learning Types!"
- **Supervised Learning:** Uses labeled data, powers ChatGPT and Grok.
  - **Explanation:** Maps inputs to outputs with guidance.
  - **Analogy:** Student with a tutor, or chef with recipe feedback.
- **Unsupervised Learning:** Finds patterns in unlabeled data, used in Meta AI clustering.
  - **Explanation:** Groups similar items autonomously.
  - **Analogy:** Child sorting toys, or librarian organizing books.
- **Reinforcement Learning:** Learns via rewards, seen in ALVINN.
  - **Explanation:** Optimizes through trial and error.
  - **Analogy:** Puppy with treats, or pilot with safe landings.
- **Class Work Sub-sub-session 2.2.1: Class Work - Classify Learning Types**
  - **Activity:** Identify learning type for given scenarios.
  - **Resources:** Scenario list.
  - **Step-by-Step Guide:**
    a. Get the scenario list (e.g., “Predicting spam with labeled data”).

b. Work in groups of 3.
c. Read each scenario and decide the type (supervised, unsupervised, reinforcement).
d. Write answers (e.g., “Spam prediction: Supervised”).
e. Discuss with the teacher.

  - **Analogy:** Categorizing game strategies.

Sub-session 2.3: Key Concepts

- **Teacher Entry:** "Having explored learning types, let’s break down ML’s building blocks. Welcome to Key Concepts!"
- **Data:** Raw information for training, like numbers or images.
  - **Explanation:** Foundation for pattern recognition.
  - **Analogy:** Pantry ingredients, or a treasure map.
- **Features:** Measurable data properties, e.g., height.
  - **Explanation:** Key inputs for learning.
  - **Analogy:** Recipe spices, or radio dials.
- **Models:** Structures learning from data.
  - **Explanation:** Generalize from examples.
  - **Analogy:** Guide dog, or custom engine.
- **Training:** Adjusts model with data.
  - **Explanation:** Minimizes errors iteratively.
  - **Analogy:** Musician practicing, or sculptor refining.
- **Class Work Sub-sub-session 2.3.1: Class Work - Define Concepts**
  - **Activity:** Define and example each concept.
  - **Resources:** Notebook.
  - **Step-by-Step Guide:**
    a. Open your notebook.
    b. Write definitions for data, features, models, training (e.g., “Data: Raw info like numbers”).
    c. Add an example for each (e.g., “Data: Weather readings”).
    d. Share one definition with a neighbor.
  - **Analogy:** Writing a recipe with examples.

Sub-session 2.4: Practical Demo

- **Teacher Entry:** "With concepts in mind, let’s apply them hands-on. Welcome to the Practical Demo!"
- **Pandas:** Python library for data manipulation.
    - **Explanation:** Loads and cleans data like spreadsheets.
    - **Analogy:** Librarian organizing shelves, or email sorter.
- **Class Work Sub-sub-session 2.4.1: Class Work - Run a Pandas Demo**
    - **Activity:** Load and explore a dataset with Pandas.
    - **Resources:** `sample_data.csv`, VS Code.
    - **Step-by-Step Guide:**
        a. Create `sample_data.csv` with:

        ```
        feature1,feature2
        1.0,2.0
        1.5,1.8
        2.0,2.2
        3.0,3.5
        3.2,3.7
        4.0,4.1
        5.0,5.2
        5.5,5.0
        6.0,6.0
        6.5,6.2
        ```

        Save in your project folder.

        b. In VS Code, open or create `PandasDemo.ipynb`, select Python kernel.

        c. In a new cell, type `import pandas as pd; data = pd.read_csv('sample_data.csv'); print(data.head())`, run with Shift+Enter (expect first 5 rows).

        d. In a new cell, type `print(data.describe())`, run, expect statistics.

        e. Save the notebook.
    - **Expected Output:** Table preview, stats (e.g., mean, std).

Session 3: Tools and Setup

Sub-session 3.1: Overview

- **Teacher Entry:** "With our demo complete, let's equip ourselves with tools. Welcome to the Overview of Tools and Setup!"
- **Python:** High-level language for AI/ML.
- **Scikit-learn:** Tools for ML tasks.
- **TensorFlow:** Framework for model building, used by Gemini.
- **Jupyter in VS Code:** Interactive coding environment.

Sub-session 3.2: Installation and Best Practices

- **Teacher Entry:** "Having reviewed tools, let's set them up properly. Welcome to Installation and Best Practices!"
- **Class Work Sub-sub-session 3.2.1: Class Work - Install Full Toolset**
    - **Activity:** Install Python, Scikit-learn, TensorFlow, and verify in VS Code.
    - **Resources:** Guide, python.org, tensorflow.org.
    - **Step-by-Step Guide:**
        a. Open terminal in VS Code, type `python --version`; if absent, go to python.org/downloads, download 3.11, install with "Add to PATH."
        b. Verify with `python --version`.
        c. Type `pip install scikit-learn` and `pip install tensorflow`, wait for "Successfully installed."
        d. In `Setup.ipynb`, add a cell, type `import sklearn; import tensorflow as tf; print(tf.__version__)`, run, expect version (e.g., 2.15).
    - **Analogy:** Stocking a kitchen.

Session 4: Resources and Activity

Sub-session 4.1: Resources

- **Teacher Entry:** "With tools ready, let's gather resources. Welcome to the Resources section!"

- **Jupyter in VS Code:** Interactive platform.
- **Sample CSV Dataset:** `sample_data.csv`.

Sub-session 4.2: Activity - "Hello, AI World!"

- **Teacher Entry:** "Equipped with resources, let's start coding. Welcome to the Activity - 'Hello, AI World!'"
- **Class Work Sub-sub-session 4.2.1: Class Work - Write and Run Hello AI Code**
  - **Activity:** Create and run a simple AI greeting script.
  - **Resources:** VS Code, `HelloAI.ipynb`.
  - **Step-by-Step Guide:**
    a. In VS Code, create `HelloAI.ipynb`, select Python kernel.
    b. In a new cell, type `name = input("Enter your name: "); print(f"Hello, {name}, welcome to AI!")`, run with Shift+Enter, enter your name (e.g., "John"), expect "Hello, John, welcome to AI!"
    c. Save the notebook.
  - **Expected Output:** Personalized greeting.
- **Class Work Sub-sub-session 4.2.2: Class Work - Modify and Share**
  - **Activity:** Modify the script and share with a partner.
  - **Resources:** `HelloAI.ipynb`.
  - **Step-by-Step Guide:**
    a. Open `HelloAI.ipynb`.
    b. Edit to `from datetime import date; name = input("Enter your name: "); print(f"Hello, {name}, l`, run, expect updated output (e.g., current date).
    c. Save changes.
    d. Show your partner and note their modification.
  - **Expected Output:** Date-included greeting.

Session 5: Reflection

Sub-session 5.1: Personal Insights

- **Teacher Entry:** "After our journey, let's reflect on our learning. Welcome to Personal

Insights!"

- **Class Work Sub-sub-session 5.1.1: Class Work - Reflect and Discuss**
  - **Activity:** Write and discuss a reflection.
  - **Resources:** Paper or doc.
  - **Step-by-Step Guide:**
    a. Take paper or open a doc.
    b. Write 3-5 sentences on a concept (e.g., “AI basics showed me pattern recognition”).
    c. Add an analogy (e.g., “Like an apprentice learning”).
    d. Pair with a classmate.
    e. Read and discuss for 5 minutes.
  - **Analogy:** Journaling after a class.